

LuciEntry HOME: An Anywhere Lucid Dream Induction Prototype

Po-Yao (Cosmos) Wang*

Exertion Games Lab, Department of
Human-Centred Computing, Monash
University
Melbourne, Australia
Po-Yao.Wang@monash.edu

Nathaniel Yung Xiang Lee*

Exertion Games Lab, Department of
Human-Centred Computing, Monash
University
Melbourne, Australia
nlee0032@student.monash.edu

Rohit Rajesh

Exertion Games Lab, Department of
Human-Centred Computing, Monash
University
Melbourne, Australia
rraj0029@student.Monash.edu

Antony Smith Loose

Exertion Games Lab, Department of
Human-Centred Computing, Monash
University
Melbourne, Australia
aloo0008@student.monash.edu

Nathan Semertzidis

Institute of Social Neuroscience
Melbourne, Australia
nathan@exertiongameslab.org

Florian ‘Floyd’ Mueller

Exertion Games Lab, Department of
Human-Centred Computing, Monash
University
Melbourne, Australia
floyd@exertiongameslab.org

ABSTRACT

Lucid dreaming is a unique state of consciousness where one is aware of dreaming whilst asleep, allowing the dreamer to control their dream content, offering various mental and physical health benefits. Existing research has utilised lab settings with manual induction techniques with researchers to induce lucid dreams, which is costly and time-consuming. Thus, there is a need for an autonomous system that would easily integrate lucid dreaming induction techniques and trigger them autonomously. In response, we present LuciEntry HOME, a portable system that utilises (1) a mobile app to guide users through pre-sleep cognitive training and (2) light, battery-powered, and wireless components for external stimulation. LuciEntry HOME emphasizes portability, autonomy and modularity, reducing the reliance on sleep labs and researchers to trigger the external stimulus manually in an effort to make lucid dreaming more accessible and facilitating future research on its application.

CCS CONCEPTS

• **Human-centered computing** → **Interaction devices.**

KEYWORDS

Lucid dreaming; induction; portable; autonomous; modular; interactive devices; prototype; system

ACM Reference Format:

Po-Yao (Cosmos) Wang, Nathaniel Yung Xiang Lee, Rohit Rajesh, Antony Smith Loose, Nathan Semertzidis, and Florian ‘Floyd’ Mueller. 2024. LuciEntry HOME: An Anywhere Lucid Dream Induction Prototype. In *Extended*

*Both authors contributed equally to this work.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3649122>

Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24), May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3613905.3649122>

1 INTRODUCTION AND RELATED WORK

Lucid dreaming is a unique state of consciousness where one is aware of dreaming whilst asleep [1]. Lucid dreamers are also aware of their ability to deliberately control the elements of their dreams [3, 8]. Lucid dreams have been associated with many benefits for both mental and physical health, including finding creative inspiration and improving problem-solving skills. It can even be used as a form of therapy to treat sleep disorders [5–7].

Recent research into lucid dreaming induction has utilised multiple lucid dreaming induction techniques to induce lucid dreams in a lab setting, such as combining pre-sleep cognitive techniques with external stimulation [2, 4]. However, lucid dreaming is a challenging task to achieve, requiring dedicated practice, and attempting to do so in a lab setting with researchers further complicates the process and affects the ability to induce lucid dreams.

2 LUCIENTRY HOME

Therefore, we designed LuciEntry HOME, an anywhere lucid dreaming induction prototype. This system utilises multiple lucid dreaming induction techniques [9] and features portable take-home components. Each component contains a specific type of stimulation to help induce lucid dreams. These prototype components are light, portable, battery-powered, and wireless, allowing users to put them anywhere in their bedroom. These prototype components are controlled by a server and work individually. The system reads a user’s brain and eye signals using Electroencephalogram (EEG) and Electrooculography (EOG) sensors, determines the sleep stage of the user with our own sleep stage algorithm, and delivers stimulation through the use of interactive technologies at the appropriate sleep stage to induce lucid dreams autonomously. While automation eases the process of lucid dream induction, the modularity of the system also allows plug-and-play of the input and output components, which gives researchers or users the flexibility of component

swapping between the desired stimuli and delivering stimuli with different combinations.

3 CONCLUSION

Through the development of LuciEntry HOME, we hope to reduce the reliance on sleep labs and researchers' manual labor for lucid dream induction. The modularity of the system allows researchers to try different combinations of external stimulation and even add newly established external stimulation that can be discovered in the future to help induce lucid dreams. With this new system, we hope to facilitate further research into lucid dreams and perhaps make lucid dreaming a new HCI research sub-field in the future.

ACKNOWLEDGMENTS

The production of the videos could not have been possible without the expertise of Mr. Aleksandar Joseski. Florian "Floyd" Mueller acknowledges the support from the Australian Research Council, especially DP190102068, DP200102612 and LP210200656. We thank Karen Konkoly and Nathan Whitmore for their guidance on lucid dream induction strategies. We also thank the Exertion Games Lab for the feedback and conceptual support they contributed toward the project and the paper.

REFERENCES

- [1] Benjamin Baird, Sergio A. Mota-Rolim, and Martin Dresler. 2019. The cognitive neuroscience of lucid dreaming. *Neuroscience & Biobehavioral Reviews* 100 (2019), 305–323. <https://doi.org/10.1016/j.neubiorev.2019.03.008>
- [2] Michelle Carr, Karen Konkoly, Remington Mallett, Christopher Edwards, Kristoffer Appel, and Mark Blagrove. 2020. Combining presleep cognitive training and REM-sleep stimulation in a laboratory morning nap for lucid dream induction. *Psychology of Consciousness: Theory, Research, and Practice* (2020).
- [3] Sophie Dyck, Michael Schredl, and Anja Kühnel. 2017. Lucid dream induction using three different cognitive methods. *International Journal of Dream Research* 10, 2 (2017), 151–156.
- [4] Daniel Erlacher, Daniel Schmid, Florian Bischof, Jennifer Hammer, and Tadas Stumbrys. 2020. Ring, ring, ring... Are you dreaming? Combining acoustic stimulation and reality testing for lucid dream induction: A sleep laboratory study. *International Journal of dream research* (2020), 267–273.
- [5] Stephen LaBerge. [n.d.]. *Benefits of Lucid Dreaming*. <https://www.altered-states.net/barry/newsletter482/>
- [6] Remington Mallett, Laura Sowin, Rachel Raider, Karen R Konkoly, and Ken A Paller. 2022. Benefits and concerns of seeking and experiencing lucid dreams: benefits are tied to successful induction and dream control. *Sleep Advances* 3, 1 (2022), zpac027.
- [7] Melanie Schädlich and Daniel Erlacher. 2012. *Applications of lucid dreams: An online study*. Universitätsbibliothek der Universität Heidelberg.
- [8] Donald W Stewart and David Koulack. 1989. A rating system for lucid dream content. *Imagination, Cognition and Personality* 9, 1 (1989), 67–74.
- [9] Shuyue Tan and Jialin Fan. 2023. A systematic review of new empirical data on lucid dream induction techniques. *Journal of Sleep Research* 32, 3 (2023), e13786. <https://doi.org/10.1111/jsr.13786> arXiv:<https://onlinelibrary.wiley.com/doi/pdf/10.1111/jsr.13786>